

DEEPALI DEWANGAN

M.Tech – Information Technology | National Institute of Technology, Raipur

EDUCATION

National Institute of Technology, Raipur — M.Tech, Information Technology | 2026 – Present

Shri Shankaracharya Technical Campus, Bhilai — B.Tech, Computer Science & Engineering | Passed: 2025
CGPA: 8.6

TECHNICAL SKILLS

Programming: Python, C/C++ **ML:** Scikit-learn, TensorFlow, Keras **Data:** NumPy, Pandas, Matplotlib
Deep Learning: ANN, CNN, Transfer Learning, U-Net **Tools:** Jupyter Notebook, VS Code, Git/GitHub

PROJECTS

Brain Tumor Detection & Segmentation System

- Developed a deep-learning-based system for brain tumor detection and classification using MRI/CT images.
- Worked with four classes: glioma, meningioma, pituitary tumor, and no tumor.
- Implemented tumor segmentation using an **EfficientNetB0-based U-Net** architecture.
- Planned a Streamlit-based interface for prediction and visualization.

Supervised Machine Learning Classification

- Implemented and compared **KNN, Logistic Regression, Naive Bayes, SVM, and Random Forest**.
- Applied preprocessing, feature scaling, model training, and performance evaluation using accuracy, precision, recall and F1-score.

Data Preprocessing & Dimensionality Reduction

- Applied missing-value handling, duplicate removal, encoding, normalization, standardization, transformation and feature selection.
- Implemented **PCA and LDA** to study dimensionality reduction, visualization and model performance.

RESEARCH INTERESTS

Large Language Models (LLMs) • Retrieval-Augmented Generation (RAG) • **RAG Collapse & Hallucination Reduction** • Machine Learning • Deep Learning • Natural Language Processing • Computer Vision

RELEVANT COURSEWORK

Machine Learning • Advanced Machine Learning • Cloud Computing • Distributed Systems • Data Structures & Algorithms • Computer Networks • Artificial Intelligence

ACHIEVEMENTS

- **GATE CSE:** Score 465 | AIR 10,341
- **M.Tech:** National Institute of Technology, Raipur — Information Technology

ACADEMIC & TECHNICAL ACTIVITIES

- Advanced Machine Learning laboratory work involving data preprocessing, supervised learning, ANN, PCA and LDA.
- Research-oriented seminar work on RAG Collapse and hallucination in Large Language Models.

LANGUAGES

English • Hindi